

ZAtop Gearless elevator machine

SM210



Description

- Permanent magnet synchronous motor with NdFeB magnets
- With ≤ 340 mm installation width for the narrowest shaft dimensions
- High efficiency
- Noise emissions: < 50 dB(A)
- Axle load: up to 4,500 kg
- Traction sheave: 240 mm to 520 mm
- Separately controllable brake circuit, type-tested as safety device against uncontrolled and unintended car movement
- Temperature monitoring by PTC or KTY thermistor

Scope of delivery

- Gearless elevator machine
- Type-tested brake system
- Traction sheave
- Absolute encoder type Heidenhain ECN1313 EnDat
- Rope guard according to EN 81

Options

- Absolute encoder type Heidenhain ECN1313 SSI or ERN1387
- Motor cable in lengths 5 m, 10 m, 15 m, 20 m and 25 m
- Mechanical hand release system for brake
- Pre-parametrised frequency inverter ZAdyn

- Motor cables
- Deflection pulleys
- ZAlift

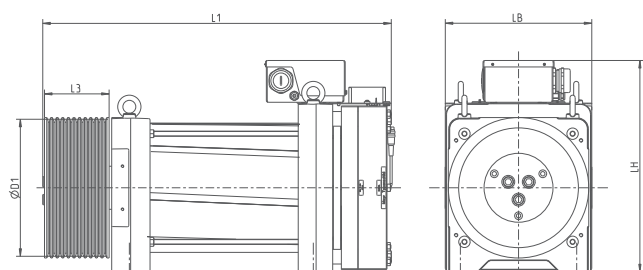
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Technical data

Motor type	Nominal torque Nm	Max. axle load kg	Nominal speed min ⁻¹	Rated output power kW
SM210.60	850	4500	60...258	5.3...23.0
SM210.70	1000			6.3...27.0

Dimensions mm

SM210.60/.70



Motor type	L1 mm	LH mm	LB mm	D1 mm	L3 mm	Weight kg
SM210.60	806	491	340	240	150	455
SM210.60				320		485
SM210.60			340	400		520
SM210.60			340	520		590
SM210.70	856	491	340	240	150	480
SM210.70				320		510
SM210.70			340	400		545
SM210.70			340	520		620

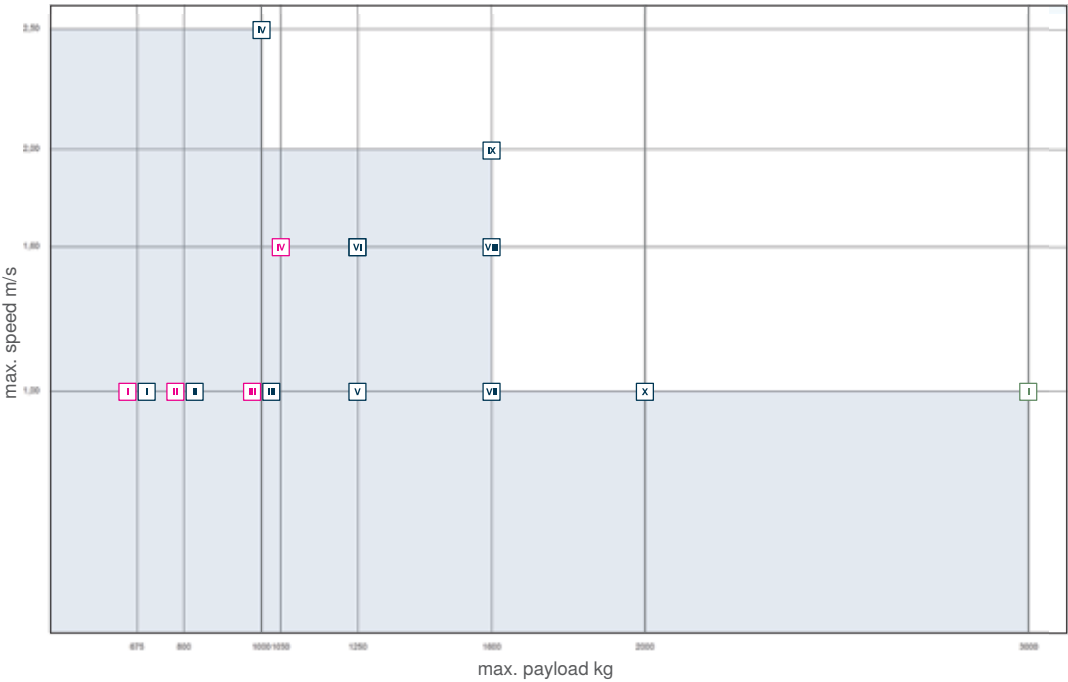


Range of possible elevator configurations for SM210

Important technical data for the drawn configuration examples are listed in the table below.
The following diagram shows the range of elevator configurations which can be achieved with the ZAtop SM210 elevator machine.
Other elevator configurations, also outside this range, are possible. Flexibility is our business!
Our calculation software ZAlift is available free of charge to you for fast and convenient calculation of your elevator project.

Example configurations

Suspension	No	Max. payload kg	Speed m/s	Motor type	Traction sheave mm	Rope Number x Ø mm	Motor power kW
1:1	I	675	1.0	SM210.60	400	4 x 10	4.5
	II	800	1.0	SM210.60	320	8 x 8	5.2
	III	1000	1.0	SM210.70	320	10 x 8	6.8
	IV	1050	1.6	SM210.70	320	10 x 8	10.9
2:1	I	675	1.0	SM210.60	520	4 x 11	5.2
	II	800	1.0	SM210.60	400	4 x 10	5.6
	III	1000	1.0	SM210.60	400	4 x 10	6.7
	IV	1000	2.5	SM210.60	520	3 x 12	15.0
	V	1250	1.0	SM210.60	320	7 x 8	8.8
	VI	1250	1.6	SM210.60	400	5 x 10	13.5
	VII	1600	1.0	SM210.60	320	10 x 8	10.8
	VIII	1600	1.6	SM210.70	400	8 x 10	17.9
	IX	1600	2.0	SM210.60	320	10 x 8	18.9
	X	2000	1.0	SM210.70	320	10 x 8	13.8
4:1	I	3000	1.0	SM210.70	400	6 x 10	18.6



Range of possible elevator configurations

Example suspension 1:1

Example suspension 2:1

Example suspension 4:1